HDL-MRCU.433

Home Control Unit

buspro

Datasheet

Issued: April 12, 2021

File Edition: A



Figure 1. Home Control Unit

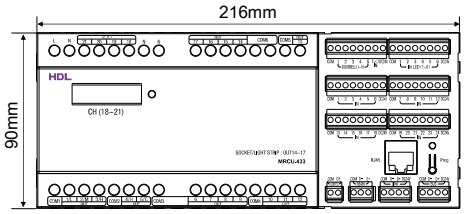


Figure 2. Dimensions - Front View

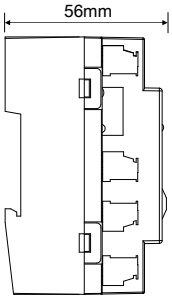


Figure 3. Dimensions - Side View

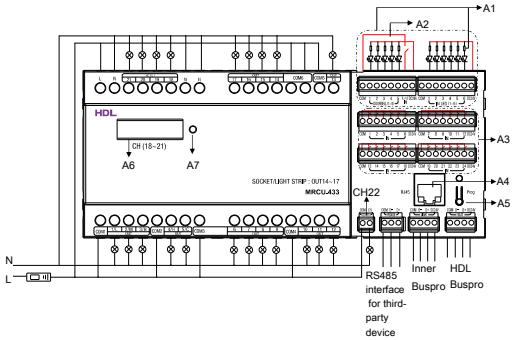


Figure 4. Wiring

Overview

Home Control Unit (See Figure 1) supports the functions of gateway, mix controller, dry contact, and LED output. The Home Control Unit has 22 channel outputs, 24-channel dry contacts and 11-channel LED outputs.

Mix control functions

- 22 zones can be set, and each zone includes 0-12 scenes.
- Channel 1 to 13 are relay outputs, and each channel outputs 5A current. The load types are recommended to be incandescent lamp, halogen lamp, low voltage halogen lamp.
- Channel 14 to 17 are relay outputs, and each channel outputs 10A current. The load types are recommended to be incandescent lamp, halogen lamp, low voltage halogen lamp.
- Channel 18 to 21 are dimming outputs, each channel outputs 0.8A current. It has short circuit protection with replaceable fuse. When the fuse is burned, the fuse indicator will flash.
- Channel 22 is 1A relay output channel.

Dry contact

- 24-channel dry contact inputs.

LED output function

- 11-channel switch indicator outputs.

RS485 function

- 1CH external RS485 Bus data receiving and sending, internally converted into HDL Buspro protocol commands.

Internet function

- Local connection control or remote connection configuration.

Components and Operation

Dimensions - See Figure 2 - 3

Wiring - See Figure 4

A1: The CH1-11 LED lamp corresponds to the dry contact CH1-11 respectively. Resistor of 1kΩ - 5kΩ recommended, which depends on the working voltage of the LED lamp.
A2: Doorbell control
A3: 24CH dry contacts
A4: RJ45 port
A5: Button and indicator: The indicator flickers in green when the module is working properly. Keep pressing for 2s to read and set the start channel, when the indicator turns red. Keep pressing for 10s, it will turn red, the module will be reset to factory settings.
Default setting parameters:
IP: 192.168.10.250
Router IP: 192.168.10.1
Port: 6000
The network segment of the module should be same as the PC.
A6: Fuse socket
A7: Indicator of fuse

ЧТУП «ОллВижн»
г. Минск, ул. Богдановича, 11
info@allvision.by hdl.by
+375 33 370 20 20
+375 29 370 20 20
ТЕХПОДДЕРЖКА: +375 29 7 511 611
©Copyright by HDL Automation Co., Ltd. All rights reserved. Specifications subject to change without notice.

Installation

Installation - See Figure 5 - 7

Step 1. Fix the DIN rail with screws.
Step 2. Buckle the bottom cap of the Home Control Unit on the edge of the DIN rail.
Step 3. Press the device on the DIN rail, slide it and fix it up until an appropriate position is adjusted.

Important Notes

- Buspro cable - CAT5E or dedicated HDL Buspro cable.
- The fuse must be aR type.
- Device must be powered off when replacing fuse.

Safety Precautions

- The installation and testing for the product must be carried out by HDL Automation Co., ltd. or its appointed service agencies. The electric construction shall comply with local laws and safety regulations.
- The device should be installed with DIN rail in DB box. HDL will not be responsible for any consequence caused by the inexpert or faulty installation and wiring methods, which are not in accordance with the instructions contained in this operating instruction.
- Please do not privately disassemble or replace any parts of the product. Otherwise, it may cause mechanical fault, electric shock, fire or personal injuries.
- Please contact our after-sales departments or our designated service agencies for your maintenance service. Product failures caused by private disassembly are not subject to this warranty.

Package Contents

HDL-MRCU.433*1 / Datasheet*1



Figure 5

1

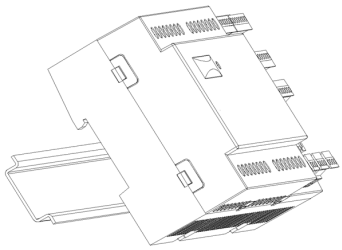


Figure 6

2

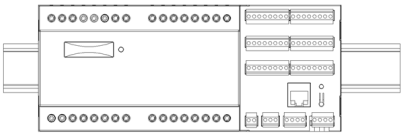


Figure 7

Figure 5 - 7. Installation

Technical Data

Basic Parameters

Working voltage	20~30V DC
Working current	300mA/24V DC
Input voltage	AC100-240V (50/60Hz)
Communication interface	HDL Buspro, RJ45
RJ45 interface	UDP/IP network interface
TRIAC	16A TRIAC, minimum load is 30W
Cable diameter of Buspro terminal	0.6-0.8mm
Relay output	5A/10A

External Environment

Working temperature	-5℃~45℃
Working relative humidity	≤90%
Storage temperature	-20℃~60℃
Storage relative humidity	≤93%

Specifications

Dimensions	216mm×90mm×56mm
Net weight	725g
Housing material	Nylon
Installation	35mm DIN rail installation (See Figure 5 - 7)
Protection rating (Compliant with EN 60529)	IP20

Name and Content of Hazardous Substances in Products

Components	Hazardous substances					
	Lead (Pb)	Mercury (Hg)	Cadmium (Cd)	Chromium VI (Cr (VI))	Poly-brominated biphenyls (PBB)	Poly-brominated diphenyl ethers (PBDE)
Plastic	o	o	o	o	o	o
Hardware	o	o	o	o	-	-
Screw	o	o	o	×	-	-
Solder	×	o	o	o	-	-
PCB	×	o	o	o	o	o
IC	o	o	o	o	×	×

The symbol “-” indicates that the hazardous substance is not contained.

The symbol "o" indicates that the content of the hazardous substances in all the homogeneous materials of the component is below the limit requirement specified in the Standard IEC62321-2015.

The symbol “×” indicates that the content of the hazardous substance in at least one of the homogeneous materials of the part exceeds the limit requirement specified in the Standard IEC62321-2015.

HDL Buspro Cable Guide

HDL Buspro	HDL Buspro Cable	CAT5/CAT5E
DATA+	Yellow	Blue/Green
DATA-	White	Blue white/Green white
COM	Black	Brown white/Orange white
24V DC	Red	Brown/Orange